

i) KCL at node V_{o1} we get

$$V_{o1} G_{m2} = V_{in} G_{m1} + \frac{V_{o1}}{r_{o1}} + V_{o1} s C_{o1}$$

$$V_{o1} G_{m2} - V_{in} G_{m1} = V_{o1} [g_{o1} + s C_{o1}]$$

$$V_{o1} = \left[\frac{V_{o1} G_{m2} - V_{in} G_{m1}}{g_{o1} + s C_{o1}} \right] \dots (1)$$

ii) KCL at node V_o

$$\frac{V_o}{r_{o2}} = -(V_o - V_{out}) s C_c$$

$$\therefore \frac{V_o}{r_{o2}} + V_o s C_c = V_{out} s C_c$$

$$\therefore V_o = \frac{V_{out} s C_c}{(g_{o2} + s C_c)} \dots (2)$$

iii) KCL at node V_{out}

$$s C_c (V_o - V_{out}) = V_{o1} G_{m2} + \frac{V_{out}}{r_{o2}} + V_{out} s C_{o2}$$

$$V_o s C_c - V_{o1} G_{m2} = V_{out} [g_{o2} + s C_{o2} + s C_c]$$

$$V_{out} = \frac{V_o s C_c - V_{o1} G_{m2}}{[g_{o2} + s C_{o2} + s C_c]} \dots (3)$$

$$V_{out} [g_{o2} + sC_c + sC_{o2}] = \frac{s^2 C_c^2 V_{out}}{(g_{oa} + sC_c)} - G_{mL} \left[\frac{V_a G_{ma} - V_{in} G_{m1}}{g_{o1} + sC_{o1}} \right]$$

$$V_{out} [g_{o2} + sC_c + sC_{o2}] - \frac{s^2 C_c^2 V_{out}}{(g_{oa} + sC_c)} = -\frac{V_a G_{ma} G_{mL}}{(g_{o1} + sC_{o1})} + \frac{V_{in} G_{m1} G_{mL}}{(g_{o1} + sC_{o1})}$$

$$V_{out} \left[(g_{o2} + sC_c + sC_{o2}) - \frac{s^2 C_c^2}{(g_{oa} + sC_c)} \right] + \frac{V_a G_{ma} G_{mL}}{(g_{o1} + sC_{o1})} = + \frac{V_{in} G_{m1} G_{mL}}{(g_{o1} + sC_{o1})}$$

$$V_{out} \left[(g_{o2} + sC_c + sC_{o2}) - \frac{s^2 C_c^2}{(g_{oa} + sC_c)} \right] + \frac{G_{ma} G_{mL} \cdot sC_c \cdot V_{out}}{(g_{o1} + sC_{o1})(g_{oa} + sC_c)} = + \frac{V_{in} G_{m1} G_{mL}}{(g_{o1} + sC_{o1})}$$

$$V_{out} \left\{ \left[(g_{o2} + sC_c + sC_{o2}) - \frac{s^2 C_c^2}{(g_{oa} + sC_c)} \right] + \frac{G_{ma} G_{mL} \cdot sC_c}{(g_{o1} + sC_{o1})(g_{oa} + sC_c)} \right\} = + \frac{V_{in} G_{m1} G_{mL}}{(g_{o1} + sC_{o1})}$$

$$\frac{V_{out}}{V_{in}} = \frac{G_{m1} G_{mL}}{(g_{o1} + sC_{o1}) \left\{ \frac{(g_{o2} + sC_c + sC_{o2})(g_{oa} + sC_c) - s^2 C_c^2}{(g_{oa} + sC_c)} + \frac{G_{ma} G_{mL} \cdot sC_c}{(g_{oa} + sC_c)(g_{o1} + sC_{o1})} \right\}}$$

$$= \frac{G_{m1} G_{mL} (g_{oa} + sC_c)}{(g_{o1} + sC_{o1}) (g_{o2} + sC_c + sC_{o2}) (g_{oa} + sC_c) - s^2 C_c^2 (g_{o1} + sC_{o1}) + G_{ma} G_{mL} sC_c}$$

$$= \frac{G_{m1} G_{mL} (g_{oa} + sC_c)}{(g_{o1} g_{oa} + sC_c g_{o1} + sC_{o1} g_{oa} + s^2 C_{o1} C_c) (g_{o2} + sC_c + sC_{o2}) - s^2 C_c^2 (g_{o1} + sC_{o1}) + G_{ma} G_{mL} sC_c}$$

$$\frac{V_{out}}{V_{in}} = \frac{G_{m1} G_{mL} (g_{oa} + s C_c)}{[g_{o1} g_{oa} g_{o2} + s C_c g_{o1} g_{o2} + s C_{o1} g_{oa} g_{o2} + s^2 C_{o1} C_c g_{o2} + s C_c g_{o1} g_{oa} + s^2 C_c^2 g_{o1} + s^2 C_{o1} C_c g_{oa} + s^3 C_{o1} C_c^2 + g_{o1} g_{oa} s C_{o2} + s^2 C_c C_{o2} g_{o1} + s^2 C_{o1} C_{o2} g_{oa} + s^3 C_{o1} C_{o2} C_c - s^2 C_c^2 g_{o1} - s^3 C_c^2 C_{o1} + G_{m1} G_{mL} s C_c]}$$

$$= \frac{G_{m1} G_{mL} (g_{oa} + s C_c)}{[g_{o1} g_{oa} g_{o2} + s C_c g_{o1} g_{o2} + s C_{o1} g_{oa} g_{o2} + s C_c g_{o1} g_{oa} + s C_{o2} g_{o1} g_{oa} + s C_c G_{m1} G_{mL} + s^2 C_{o1} C_c g_{o2} + s^2 C_{o1} C_c g_{oa} + s^2 C_c C_{o2} g_{o1} + s^2 C_{o1} C_{o2} g_{oa} + s^3 C_{o1} C_{o2} C_c]}$$

$$= \frac{G_{m1} G_{mL} (g_{oa} + s C_c)}{g_{o1} g_{o2} g_{oa} + s [g_{o1} g_{o2} C_c + C_{o1} g_{oa} g_{o2} + g_{o1} g_{oa} C_c + C_{o2} g_{o1} g_{oa} + C_c G_{m1} G_{mL}] + s^2 [C_{o1} C_c g_{o2} + C_{o1} C_c g_{oa} + C_c C_{o2} g_{o1} + C_{o1} C_{o2} g_{oa}] + s^3 (C_{o1} C_{o2} C_c)}$$

$$= \frac{g_{oa} [G_{m1} G_{mL} (1 + s C_c / g_{oa})]}{g_{o1} g_{o2} g_{oa} \left\{ 1 + s \left[\frac{C_c}{g_{oa}} + \frac{C_{o1}}{g_{o1}} + \frac{C_c}{g_{o2}} + \frac{C_{o2}}{g_{o2}} + \frac{C_c G_{m1} G_{mL}}{g_{o1} g_{o2} g_{oa}} \right] + s^2 \left[\frac{C_{o1} C_c}{g_{oa} g_{o1}} + \frac{C_{o1} C_c}{g_{o1} g_{o2}} + \frac{C_c C_{o2}}{g_{o2} g_{oa}} + \frac{C_{o1} C_{o2}}{g_{o1} g_{o2}} \right] + s^3 \left[\frac{C_{o1} C_{o2} C_c}{g_{o1} g_{o2} g_{oa}} \right] \right\}}$$

$$\text{Now } s^3 [C_{o1} C_{o2} C_c / g_{o1} g_{o2} g_{oa}] \ll 1$$

$$\therefore \frac{V_{out}}{V_{in}} = \frac{g_{oa} [G_{m1} G_{mL} (1 + sC_c/g_{oa})]}{g_{o1} g_{o2} g_{oa} \left[1 + s \left[\frac{C_c}{g_{oa}} + \frac{C_{o1}}{g_{o1}} + \frac{C_c}{g_{o2}} + \frac{C_{o2}}{g_{o2}} + \frac{C_c G_{m1} G_{mL}}{g_{o1} g_{o2} g_{oa}} \right] + s^2 \left[\frac{C_{o1} C_c}{g_{o1} g_{oa}} + \frac{C_{o1} C_c}{g_{o1} g_{o2}} + \frac{C_c C_{o2}}{g_{o2} g_{oa}} + \frac{C_{o1} C_{o2}}{g_{o1} g_{o2}} \right] \right]}$$

$$= \frac{G_{m1} G_{mL} (1 + sC_c/g_{oa})}{g_{o1} g_{o2}} \cdot \frac{1 + s \left[\frac{C_c}{g_{oa}} + \frac{C_{o1}}{g_{o1}} + \frac{C_c}{g_{o2}} + \frac{C_{o2}}{g_{o2}} + \frac{C_c G_{m1} G_{mL}}{g_{o1} g_{o2} g_{oa}} \right] + s^2 \left[\frac{C_{o1} C_c}{g_{o1} g_{oa}} + \frac{C_{o1} C_c}{g_{o1} g_{o2}} + \frac{C_c C_{o2}}{g_{o2} g_{oa}} + \frac{C_{o1} C_{o2}}{g_{o1} g_{o2}} \right]}{1}$$

$$\text{Now } g_{oa} = \frac{1}{r_{oa}} = \frac{1}{1/G_{ma}} = G_{ma}$$

$$g_{o1} = 1/r_{o1}$$

$$g_{o2} = 1/r_{o2}$$

$$= \frac{G_{m1} G_{mL} r_{o1} r_{o2} (1 + sC_c/G_{ma})}{1 + s \left[G_{ma} C_c + C_{o1} r_{o1} + C_c r_{o2} + C_{o2} r_{o2} + C_c G_{mL} r_{o1} r_{o2} \right] + s^2 \left[C_{o1} C_c r_{o1} G_{ma} + C_{o1} C_c r_{o1} r_{o2} + C_c C_{o2} r_{o2} G_{ma} + C_{o1} C_{o2} r_{o1} r_{o2} \right]}$$

$\therefore G_{ma}$ is of the order of 10^{-6} and Caps are of the order of 10^{-12} compared to r_{o1} terms which are of the order of 10^3 to 10^6 .
 $\therefore$ ignoring smaller terms from denominator we get

$$\frac{V_{out}}{V_{in}} = \frac{G_{m1} G_{mL} r_{o1} r_{o2} (1 + sC_c/G_{ma})}{1 + s [C_{o1} r_{o1} + C_c r_{o2} + C_{o2} r_{o2} + C_c G_{mL} r_{o1} r_{o2}] + s^2 [C_{o1} C_c r_{o1} r_{o2} + C_{o1} C_{o2} r_{o1} r_{o2}]}$$

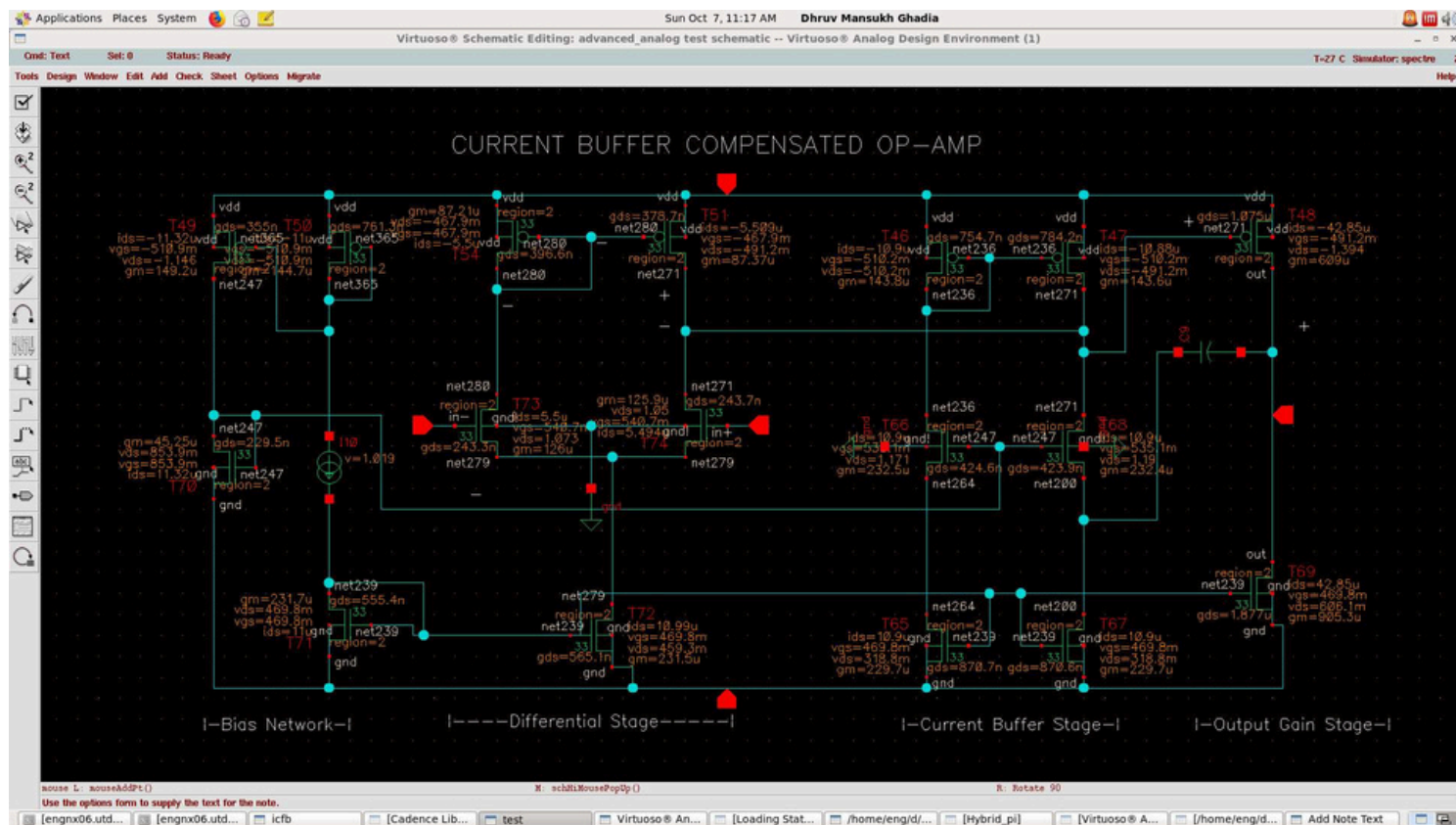
$$\therefore \frac{V_{out}}{V_{in}} \approx \frac{G_{m1} G_{mL} r_{o1} r_{o2} (1 + sC_c/G_{ma})}{1 + s (C_{o1} r_{o1} + C_c r_{o2} + C_{o2} r_{o2} + C_c G_{mL} r_{o1} r_{o2}) + s^2 (C_{o1} C_c r_{o1} r_{o2} + C_{o1} C_{o2} r_{o1} r_{o2})}$$

$$\therefore \text{we get } p_1 \approx - \frac{1}{G_{mL} r_{o1} r_{o2} C_c}$$

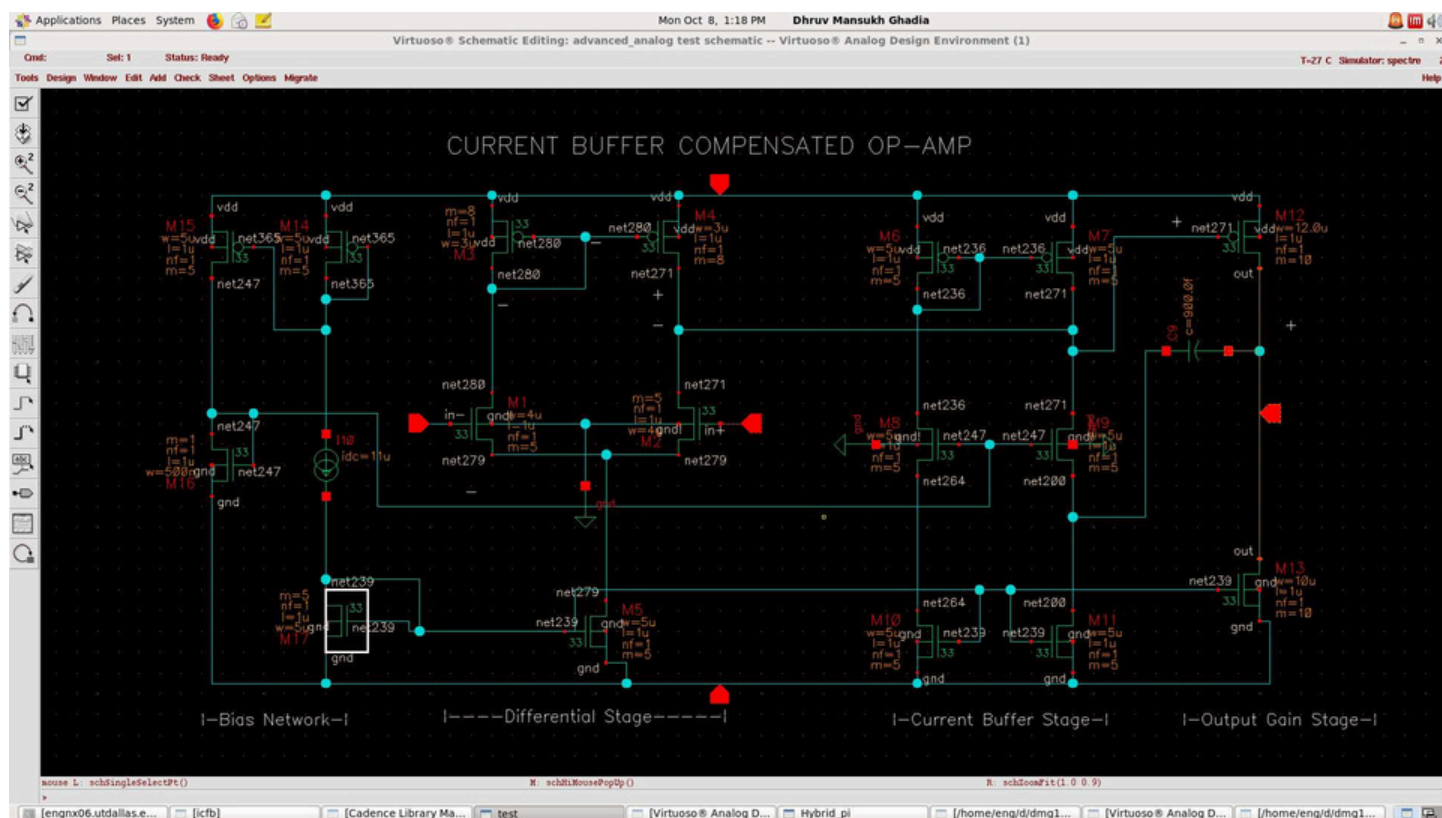
$$Z_1 \approx - \frac{G_{ma}}{C_c}$$

Current Buffer Compensated Op-amp Design and Simulation Circuit

Schematic with DC Operating point:



Circuit Schematic with Transistor Sizes:



TRANSISTOR SIZE SUMMARY:

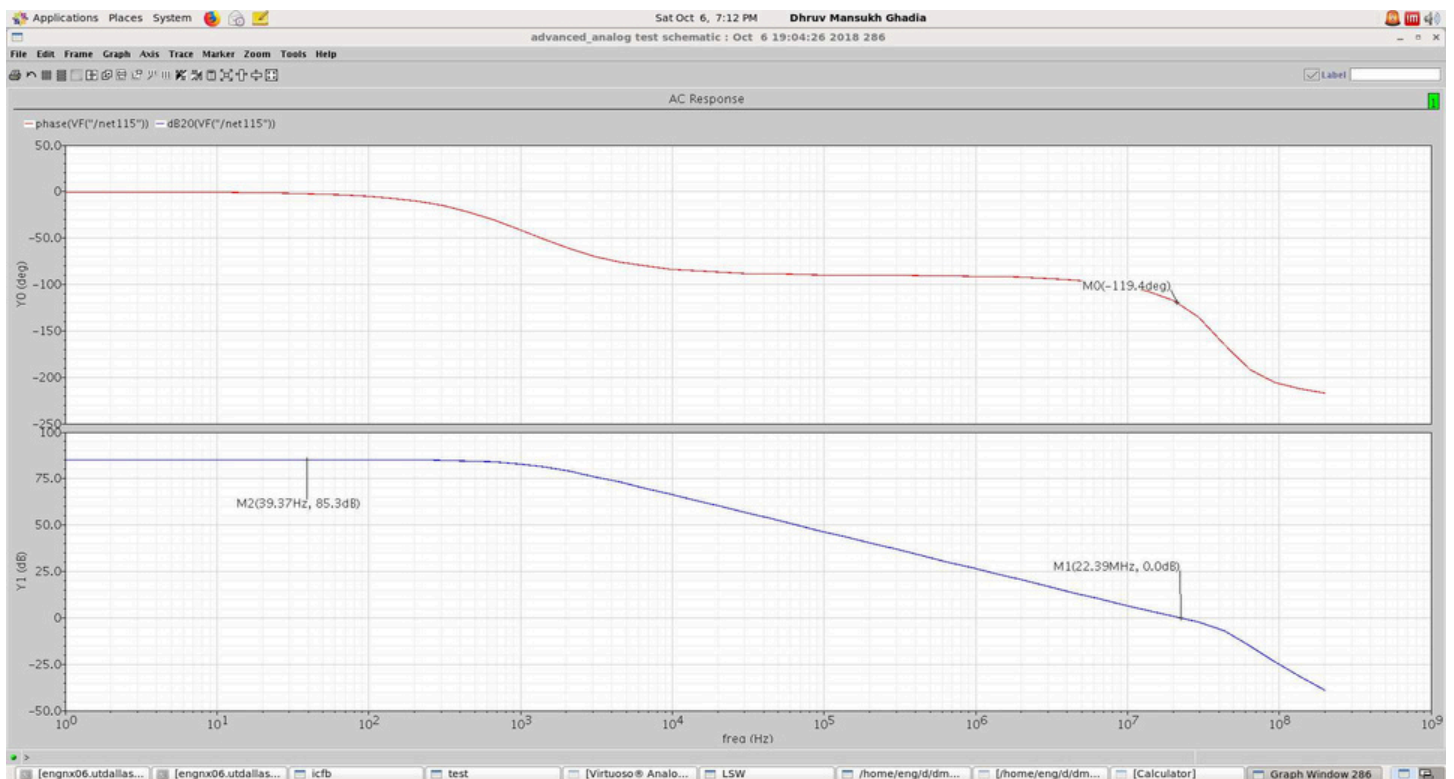
Transistor	Width/Length	Multiplicity	Network
M1	4u/1u	5	Differential Input Stage
M2	4u/1u	5	
M3	3u/1u	8	
M4	3u/1u	8	
M5	5u/1u	5	
M6	5u/1u	5	Current Buffer Stage
M7	5u/1u	5	
M8	5u/1u	5	
M9	5u/1u	5	
M10	5u/1u	5	
M11	5u/1u	5	Output Stage
M12	12u/1u	10	
M13	10u/1u	10	Bias Network
M14	5u/1u	5	
M15	5u/1u 0.5u/1u	5	
M16	5u/1u	1	
M17		5	

GAIN AND PHASE:

Unity Gain Frequency Achieved = 22.33MHz

Phase Margin Achieved = 60.6°

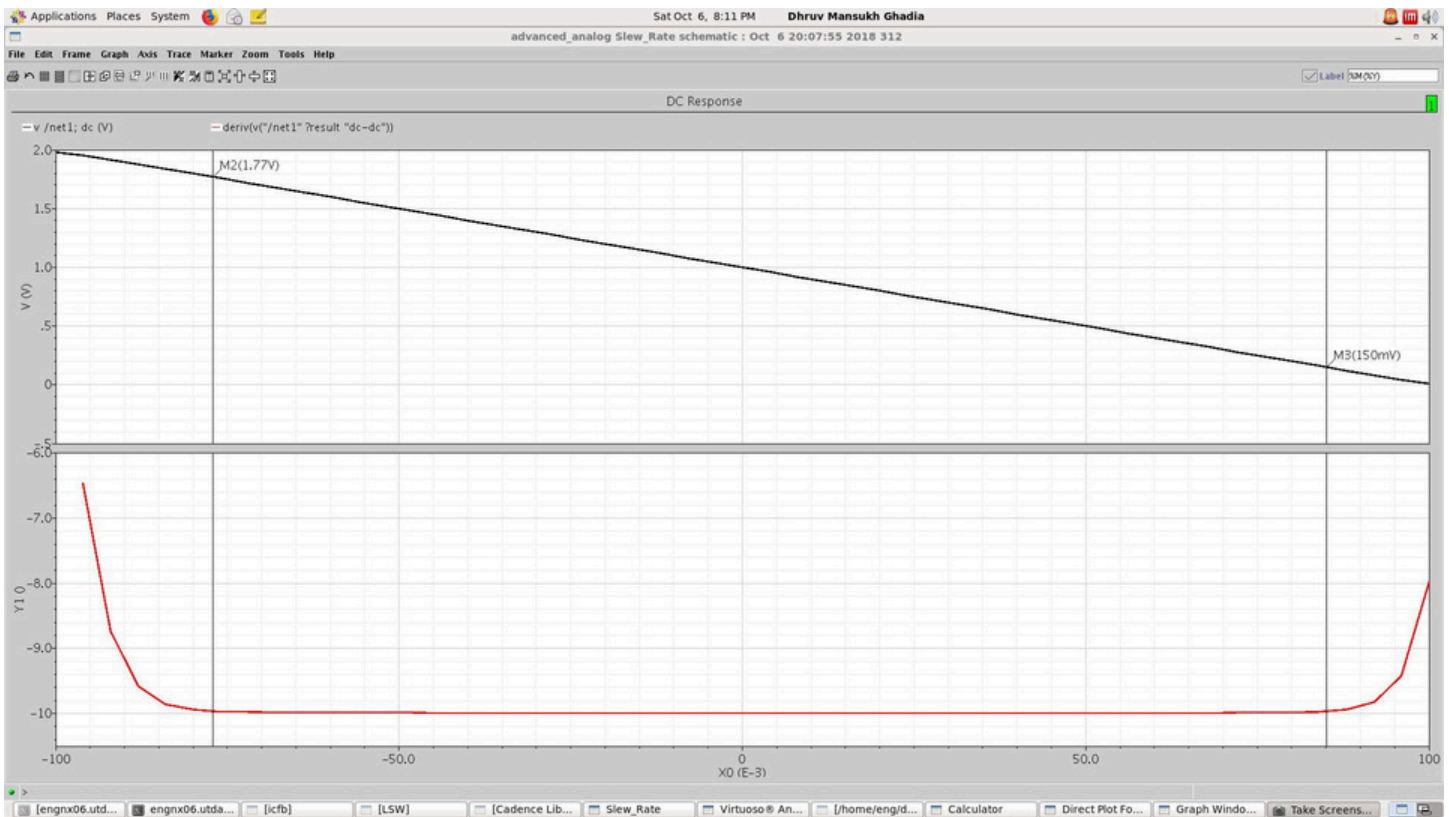
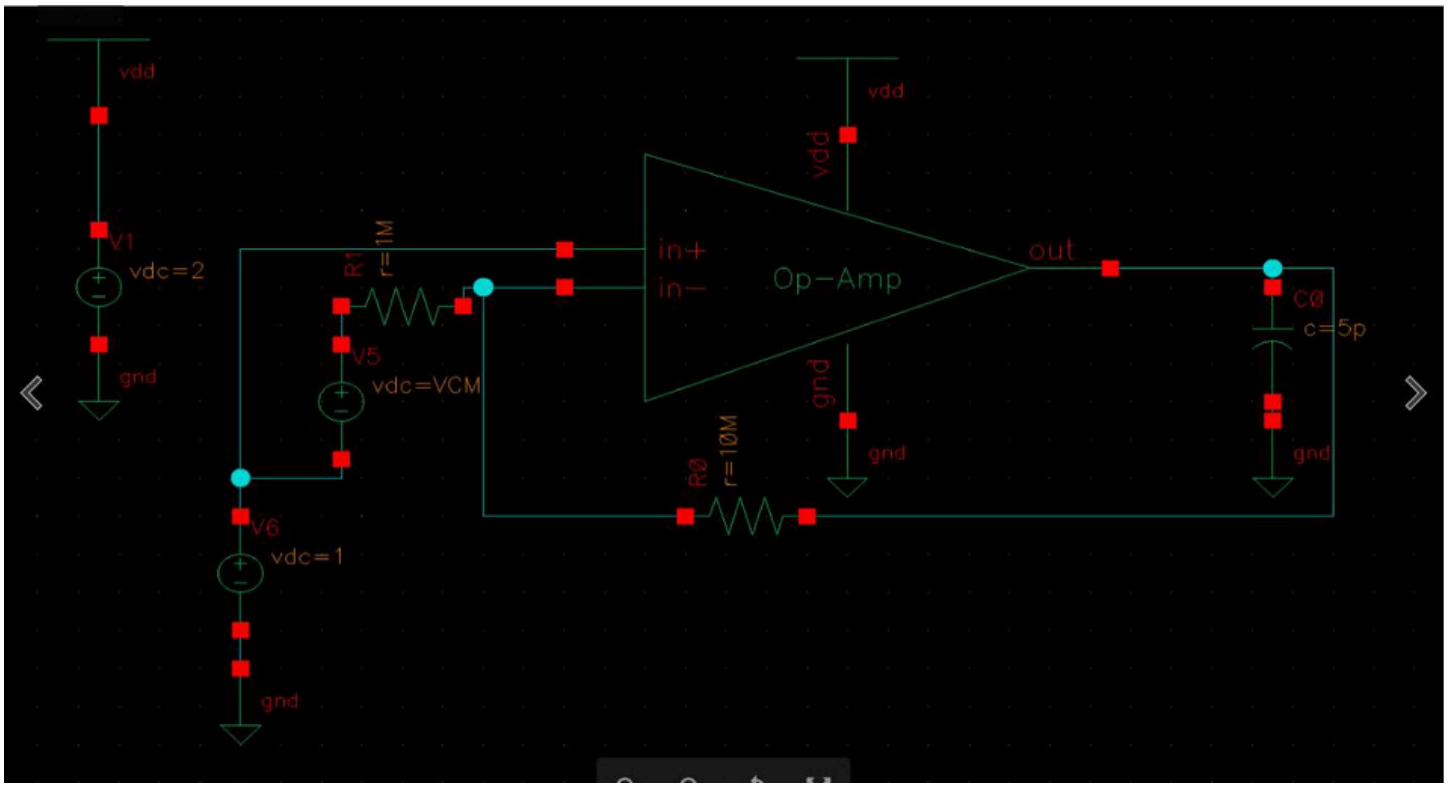
DC Gain = 85.3 dB



OUTPUT VOLTAGE SWING RANGE:

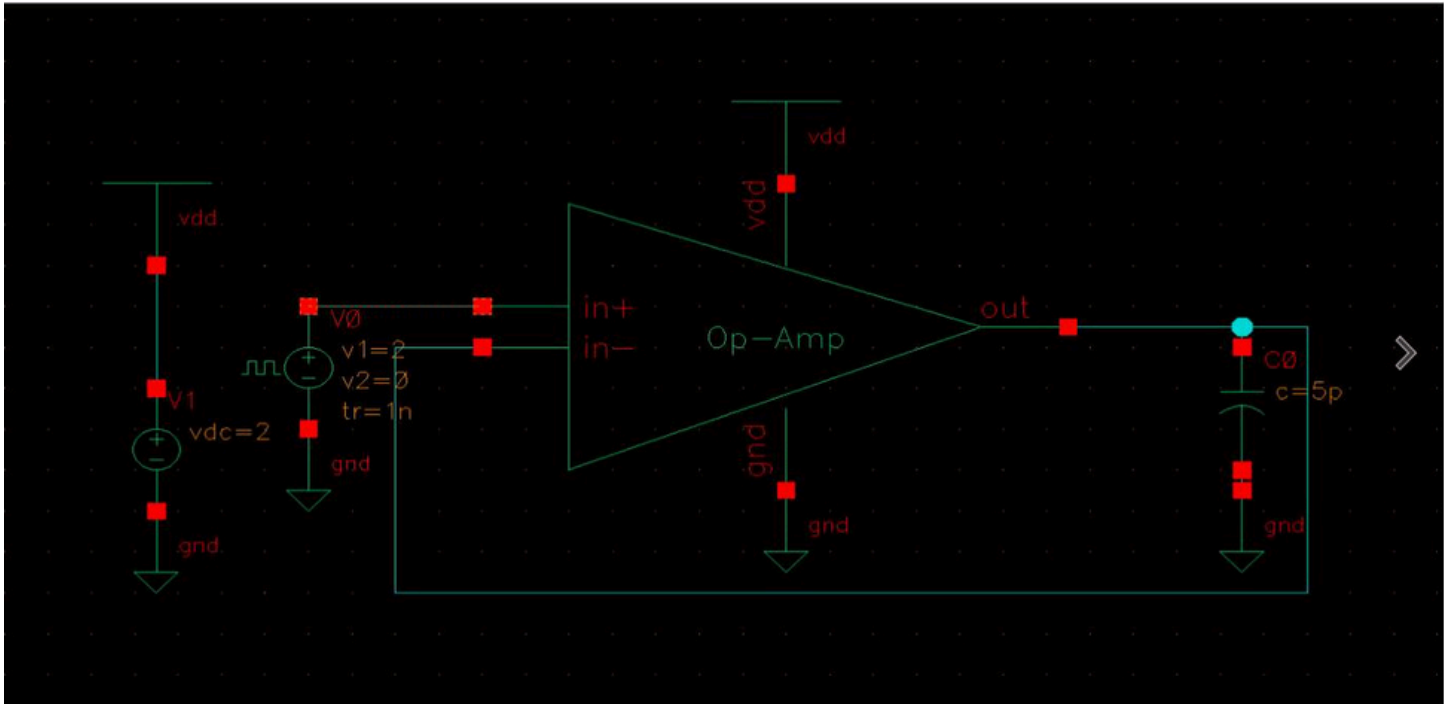
$V_{o(\min)} = 0.146V$ $V_{o(\max)} = 1.8V$

Test Bench:

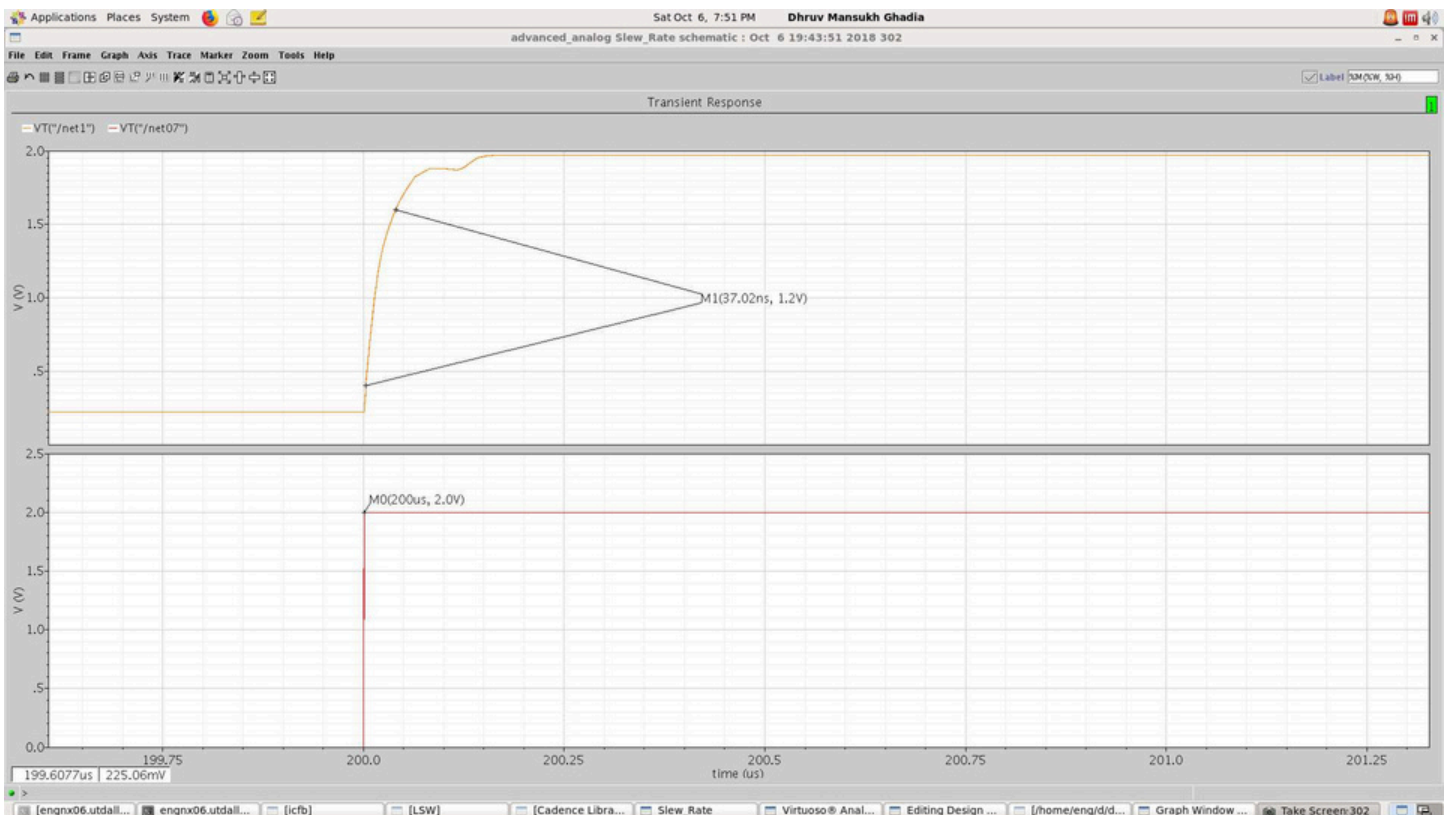


SLEW RATE:

Test Bench:

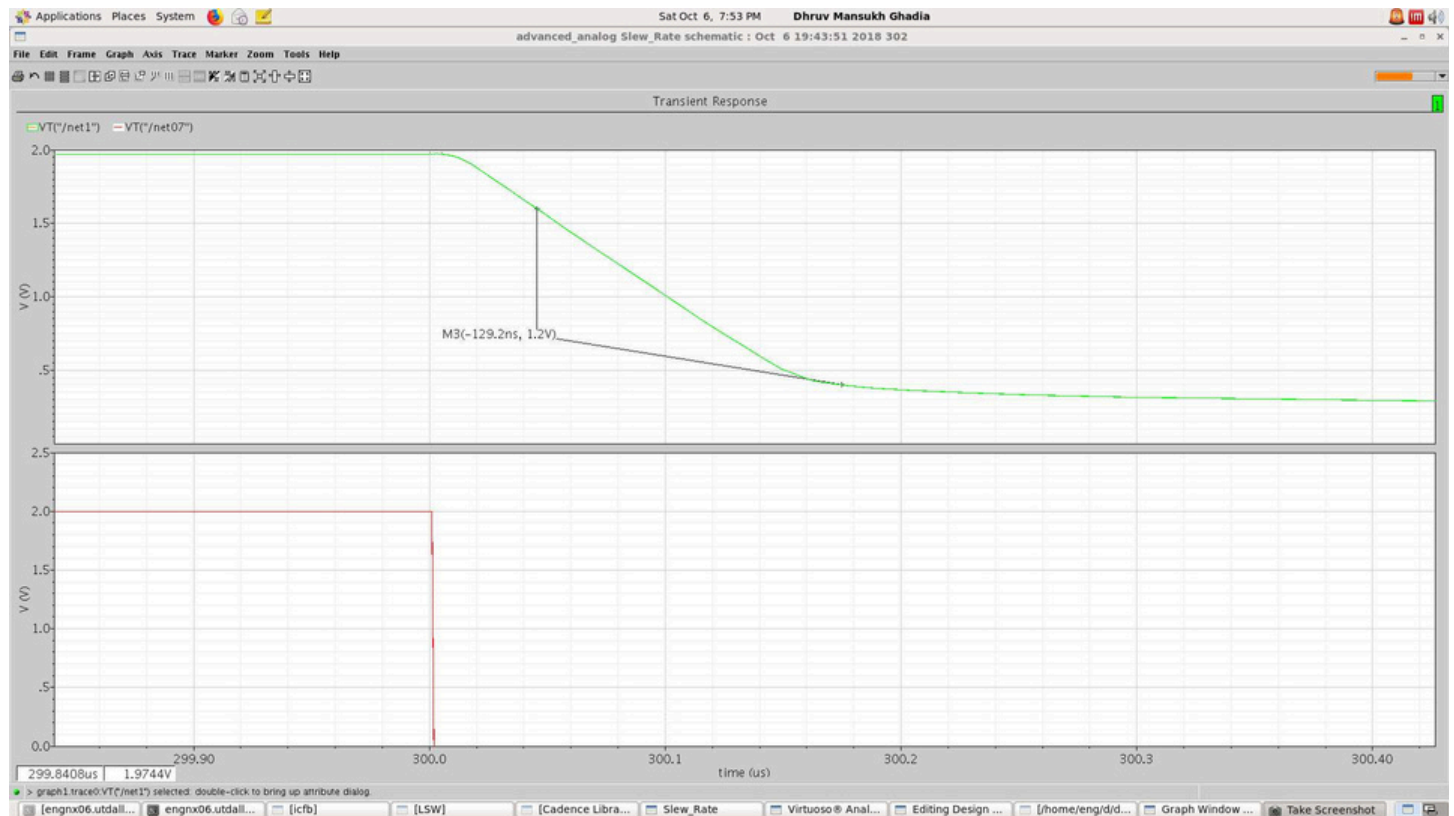


Positive Slew Rate = 32.4V/us



Negative Slew Rate = $9.28\text{V}/\mu\text{s}$

Total Slew rate = $(9.28+32.4)/2 = 20.84\text{ V}/\mu\text{s}$



TOTAL POWER CONSUMPTION (Including Bias network):

Window Expressions Info	
signal	OP ("V1" "??")
i	-98.66u
pwr	-197.3u
v	2

Total Current without including Bias Network = $11\mu\text{A} + 11\mu\text{A} + 11\mu\text{A} + 42\mu\text{A} = 75\mu\text{A}$

Power Consumption = $2\text{V} \times 75\mu\text{A} = 150\mu\text{W}$

HYBRID-PI MODEL SIMULATION:

For Hybrid-pi

$$\text{For } 60^\circ \text{ PM} \quad C_c = 0.22 C_L$$

$$C_L = 5 \text{ pF}$$

$$\therefore C_c = 0.22 \times 5 \text{ pF}$$

$$C_c = 1.1 \text{ pF}$$

$$\begin{aligned} \text{Now } G_{m1} &= U_{GB} \times C_c \\ &= 2\pi \times 20 \times 10^6 \times 1.1 \text{ p} \end{aligned}$$

$$G_{m1} = 138.23 \mu$$

$$G_{mL} \approx 10 G_{m1}$$

$$\approx 10 \times 138.23 \mu$$

$$G_{mL} \approx 1.38 \text{ m}$$

$$\text{Assume } \lambda = 0.25$$

$$\text{Slew Rate} = \frac{I_5}{C_c} \Rightarrow I_5 = 2.2 \mu\text{A}$$

$$r_{ds} = \frac{1}{\lambda I_D} = \frac{1}{2 \times 0.25 \times 2.2} = 1.6 \text{ M}\Omega$$

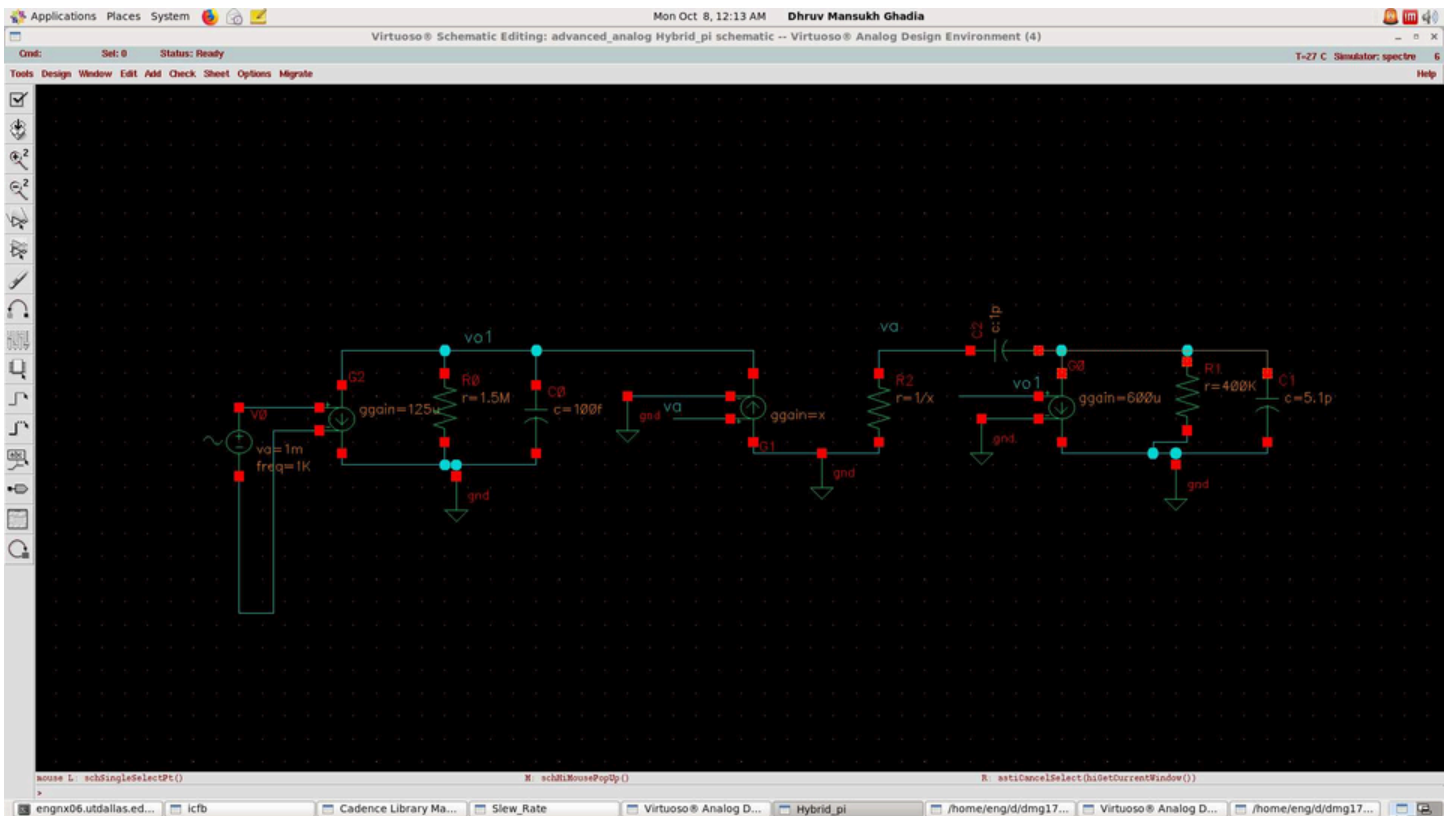
$$C_2 \approx C_L = 5 \text{ pF}$$

$$\text{Taking } C_1 = 100 \text{ f}$$

$$\text{For } G_{m1} \ll \frac{G_{m2} C_c^2}{C_{01}(C_{02} + C_c)} = \frac{1.38 \text{ m} \times 1.1}{(0.1)(1.1 + 5)} \ll 2.3 \text{ m}$$

$$\therefore \text{Taking } G_{m1} = 200 \mu$$

$$r_{OA} = \frac{1}{G_{m1}} = 5 \text{ K}\Omega$$



Virtuoso® Analog Design Environment (4)

Status: Ready T=27 C Simulator: spectre 96

Session Setup Analyses Variables Outputs Simulation Results Tools Help

Design		Analyses			
Library	advanced_analog	#	Type	Arguments...	Enable
Cell	Hybrid_pi	1	ac	1 100M Auto.. Star..	yes
View	schematic				

Design Variables			Outputs			
#	Name	Value	#	Name/Signal/Expr	Value	Plot Save March
1	x	204u	1	GBW	20.71M	
			2	PM	63.83	

Plotting mode: Replace

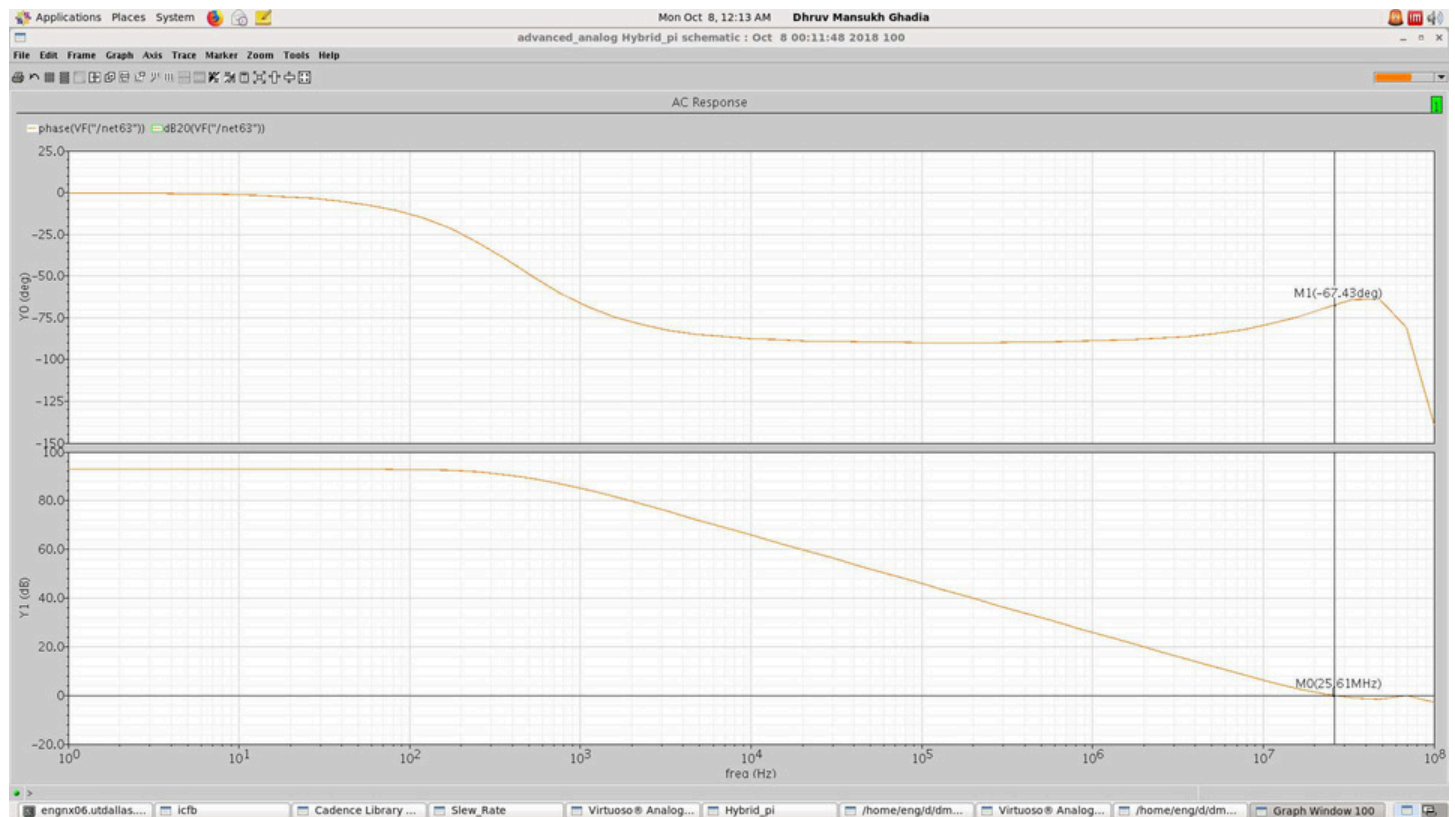
> Results in /home/eng/d/dmg170230/simulation/Hybrid_pi/spectre/schematic

AC
 TRAN
 DC
 X
 Y
 Z
 3D
 Pencil
 Traffic light
 Traffic light
 Waveform

Gain and Phase Margin(from Hybrid Pi):

GBW = 25.61MHz

PM = 63.83



DESIGN SPECIFICATIONS SUMMARY:

Parameter	Target Specifications	Achieved Specifications
Differential voltage gain	> 80dB	85.3 dB
Output voltage swing range	Vomin: 0.15V Vomax: 0.85V	Vomin: 0.15 V Vomax: 1.7 V
Phase Margin: f(GB)	>55°	60.6°
Unity Gain-bandwidth: GB	> 20MHz	22.3 MHz
Slew rate:	>2 V/us	20.84 V/us
Power dissipation	<200 uW	190 uW

Q2-a,b]

Q2] a)

$$C_1 = 8$$

$$C_2 = 3.75$$

C_u = unit cap of size $4\mu m \times 2\mu m$

ie $x_u = 4\mu m$

Now Let $C_1 = 8C_u$

and $C_2 = 2C_u + 1.75 C_{nu}$

$$1+f = 1.75$$

$$\therefore C_2 = I_2 C_u + (1+f) C_{nu}$$

Size of non-unit Capacitance

$$C_{nu} = (1+f) C_u$$

$$C_{nu} = 1.75 C_u$$

$$N = 1+f = 1.75$$

$$\therefore y_{nu} = x_u \left(N - \sqrt{N^2 - N} \right)$$

$$= 4\mu m \left(1.75 - \sqrt{1.75^2 - 1.75} \right)$$

$$y_{nu} = 2.417\mu m$$

$$\therefore x_{nu} = \frac{N x_u^2}{y_{nu}} = \frac{1.75 \times 16}{2.417} = 11.58\mu m$$

$$\text{we want } \frac{P_{nu}}{A_{nu}} = \frac{P_u}{A_u}$$

$$\therefore P_{nu} = (2.417 + 11.58) \times 2 = 27.994\mu m$$

$$A_{nu} = 2.417 \times 11.58 = 27.9886\mu m^2$$

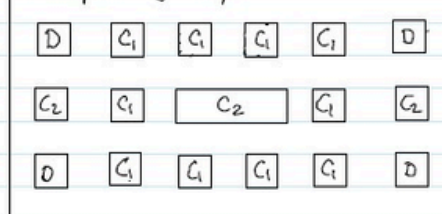
$$P_u = 16\mu m$$

$$A_u = 16\mu m^2$$

$$\therefore \frac{P_{nu}}{A_{nu}} \approx \frac{P_u}{A_u} \approx 1$$

$$\text{Ideally } \frac{P_{nu}}{A_{nu}} = \frac{P_u}{A_u} = 1$$

Floorplan of Cap:



where :

$$\square = C_u$$

$$\square = C_{nu}$$

$\square$ = Dummy Caps of unit size

Rounding off to nearest $0.1\mu m$

$$x_{nu} = 11.5\mu m \quad \& \quad y_{nu} = 2.4\mu m$$

$$P'_{nu} = 2 \times (11.5 + 2.4) = 27.8$$

$$A'_{nu} = 11.5 \times 2.4 = 27.6$$

$$\text{Actual Ratio : } \frac{P'_{nu}}{A'_{nu}} = \frac{27.8}{27.6} = 1.00724$$

$$\% \text{ error} = \frac{1.00724 - 1}{1} \times 100$$

$$\% \text{ error} = 0.72\%$$